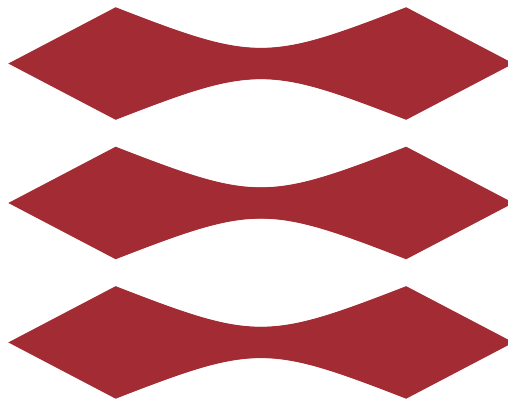


DTU



Advanced Business Analytics - Final Project

B41 Bus Route Delay Forecasting

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EXECUTIVE SUMMARY

B41 Bus Route Delay Forecasting • Non-Technical Audience

348K

GPS records analysed

4 mo.

Data window 2017

1 in 4

Time slots severely
delayed

80%

Disruptions persist
≥ 30 min

6 h

Forecast horizon

56%

Slots with nearby
events

3

Approaches tested

8.7'

Average delay

Why This Matters

Bus delays on the B41 are not short-lived inconveniences. Once a disruption starts, there is an 80% chance it is still active 30 minutes later, and it typically takes over three hours before conditions normalise. For operators managing one of Brooklyn's busiest routes, this means a single incident can cascade through the afternoon or evening schedule with no early warning. Today, disruptions are discovered only after passengers are already affected. Our goal is to change that: a forecasting system that flags likely disruptions up to six hours in advance, giving operators time to re-route buses, adjust staffing, and inform passengers before a delay fully develops.

The Data

We drew on four months of official MTA bus tracking data from 2017, spanning summer, autumn, and winter to capture different delay conditions. To build each forecast, we combined the route's recent delay history with weather at the time, nearby public event schedules, and time-of-day patterns such as rush hour. All results reported here come from December data the model never processed during development, so performance figures reflect genuine out-of-sample predictions, not a good fit to data the system already knew.

Key Findings

- Short-term predictions are reliable enough to act on. At 30 minutes ahead, the model predicts average delay to within 1.6 minutes and raises a disruption alert that is correct 9 times out of 10. At one hour ahead it is still accurate to within 2.2 minutes.
- Forecasting with simple methods adds genuine value only within the first two hours. Up to that point the model is up to 45% more accurate than a historical average. Beyond two hours, the historical average becomes just as good.
- The LSTM based model performs better than all baseline models on most metrics for regression and disturbance prediction.
- The uncertainty model correctly flags significant disruptions >50% of the time.

Recommendations

- 1 **Deploy a real-time disruption alert.** A 30–90 minute warning system is technically ready. Control centers could re-route buses and notify passengers before a delay fully develops.
- 2 **Use historical patterns for shift planning.** For decisions 3–6 hours out, staffing, bus allocation, the most reliable guide is knowing what delays typically look like at that time of day and day of week.
- 3 **Treat the events calendar as operational intelligence.** Public events correlate strongly with delays. Operators should cross-reference the events calendar when planning service on days with large gatherings near the route.
- 4 **Invest in richer data.** The most impactful improvements would come from two sources: recording actual travel times between stops — data not currently available in the MTA feed — and expanding the analysis to a full year.

LIMITATIONS

ONLY 4 MONTHS OF DATA, AND NOT CONSECUTIVE

We have June, August, October and December 2017, with gaps in between. A continuous year of data would make the analysis significantly more robust. Furthermore, the data used for training (2/3 first months) has not the same distribution as the validation and test (last 1/2 months) data.

NO ACTUAL TRAVEL TIMES BETWEEN STOPS

The MTA feed does not record how long each bus takes between stops. We had to reconstruct this from position snapshots, introducing imprecision.

EVENTS TREATED AS PRESENT OR ABSENT

We know whether a public event was happening near a stop, but not its size or type. A sellout concert and a small street fair receive the same weight.

ANALYSIS COVERS THE WHOLE ROUTE, NOT SECTIONS

Delays are averaged across the entire B41. Disruptions often start in one location, pinpointing that would allow a more targeted response.

SINGLE-ROUTE SCOPE

Results are specific to the B41 route. Deploying the approach on other routes requires refitting the QR layer with route-specific LSTM outputs and feature sets.

Highlights

B41 Bus Route Delay Forecasting, aspects we are proud of

1. NEAR-TERM DISRUPTION DETECTION

At 30 minutes ahead, the model catches 8 in 10 developing disruptions with a false-alarm rate under 10% — meaning operators can trust that when an alert fires. Precision stays above 80% through the first 90 minutes, giving a reliable early-warning window before confidence degrades.

2. EVENT DATA INTEGRATION

Public-event binary flags at 5 specific stops along the B41 route are included as predictors. This signal, active in over 56% of all windows, is rarely leveraged in transit forecasting and emerged as a meaningful driver at longer horizons.

3. DISRUPTION PERSISTENCE ANALYSIS

We quantified how long disruptions last once they start: 80% probability of persisting 30+ minutes, 71% at 60 minutes. This result directly and intuitively justifies the 6-hour forecast horizon to a non-technical audience without requiring any statistical background.

4. TWO-HOUR FORECASTING BOUNDARY

The baseline analysis revealed that the regression model's advantage over a simple historical average disappears at around two hours ahead. At 30 minutes, the model is 45% more accurate, by 2.5 hours, the historical average actually performs better. This boundary shapes the entire system design: the real-time alert layer is built for the first two hours, and long-range planning relies on structural patterns.

5. FEATURE IMPORTANCE SHIFT ACROSS HORIZONS

Ridge regression coefficients reveal a clear transition: lag features dominate at short horizons and calendar/event features take over at long horizons. This insight directly guides feature selection for future models.

6. RELEVANT TRAINING DATA FOR THE LSTM ARCHITECTURE

The training data for the LSTM features weather, events, time and delay data. Specifically, it features information such as day of the week and daily/weekly lags to capture seasonal behaviours along with disturbance information from events, holidays and precipitation data for example.

7. A VERSATILE LSTM ARCHITECTURE

The architecture of the Sequence to Sequence LSTM model enables to capture context from past information while taking into account exogenous data from both past information and available future predictions. The adaptive parameters L and n are easily editable and provide hyperparameters to play with during training.

8. BETTER FOUNDATIONS LEAD TO SHARPER PREDICTIONS: LSTM VS. RIDGE.

By developing a more advanced deep learning model, we achieved a **high reliability rate**, where in 4 out of 5 (**80%**) cases the actual delay fell within the predicted range. This is **more than double** the reference model's reliability rate of **40.9%**.

9. A MODEL THAT KNOWS WHEN IT IS WRONG: CALIBRATED UNCERTAINTY.

A two-step improvement process was applied that wraps each forecast in an honest uncertainty range. Applied to the advanced model, this process narrowed the uncertainty range from 14.3 to 12.8 minutes while maintaining the 80% coverage target. The same approach was validated on the reference model, confirming it works regardless of the underlying prediction method.